Plotraa handover: dealer-chosen sign blur

For: Sant (review, merge, deploy) | From: Pankaj's session with Claude | Date: 19 Oct 2026

Branch: claude/sign-blur-areas in BOTH repos. **Stacked on the street-view PRs** (backend #62, frontend #44): merge those first, then these.

In one line: once the location is final, the dealer can mark signboards or the house number on the front of the property. Those areas are blurred on the live street view, and the blur follows the house as buyers look around. Everything else stays sharp.

1. How a dealer uses it

- After the pin is right, the dealer's preview page shows a new section: **Hide signboards or the house number (optional)**, with the same locked street view buyers will get.
- Tap **+ Hide an area**: a box appears in the middle of the view. Drag it over the sign; drag its orange corner to resize. Add up to 8 boxes; **Remove selected** deletes one.
- Tap **Save hidden areas**. Nothing is hidden unless the dealer does this. The step is optional.
- When the dealer approves ("haan"), the live page blurs exactly those areas. After that they cannot be changed (same rule as the pin).
- If the dealer later moves the pin, the street view changes, so the saved areas are ignored and the dealer marks them again.

2. How it works

Part	What it does	Where
Storage	New column listings.front_blur (jsonb, additive migration with down): { panoId, regions: [{ heading, pitch, width, height }] } in degrees, on the panorama buyers see.	migrations/20261019_01_listing_front_blur.js
Save	PATCH /api/v1/public/listings/:slug/blur-regions. Pre-approval only (409 once live). The panoId sent must equal the current panorama or it is rejected as a stale view. At most 8 areas; malformed ones are dropped.	publicListingController.js, frontViewService.js (sanitizeBlurRegions)
Read	The public listing returns streetView: { panoId, heading, blurRegions } for pre-approval and live listings. Areas whose panoId no longer matches are returned empty.	publicListingController.js
Blur	Frontend draws a blurred box (CSS backdrop-filter) at each area. Angles are converted to screen pixels from the current heading, tilt, zoom and size, so it tracks the sign. Live boxes are padded 12% so the whole sign is covered.	SignBlurLayer.jsx, utils/streetViewProjection.js
Editor	Same component with editing on: dashed boxes, drag to move, corner to resize.	PropertyView.jsx, PropertyMapMedia.jsx

3. IMPORTANT: verify on a real listing before relying on it

The blur boxes are positioned with a model of how Google's street-view viewer maps zoom to field of view (90 degrees wide at zoom 1, focal length doubling per zoom level). It was tested in a headless browser against a **stand-in** for Google Maps (the box moves left when turning right and grows when zooming), **not against real Google imagery**. If the model is slightly off, a box can drift from the sign at some zoom levels. Check this on a real listing with a visible sign before telling dealers to rely on it.

- Mark a sign at the default zoom, save, open the live page, then zoom in and out and turn left and right: the box should stay on the sign.
- If it drifts with zoom, adjust FOCAL_AT_ZOOM_1 at the top of src/utils/streetViewProjection.js (0.5 = 90 degree field of view at zoom 1). This is the only place to tune.
- The 12% padding (LIVE_PADDING in SignBlurLayer.jsx) is the safety margin for small errors.
- Blur is a browser overlay: someone using developer tools could remove it. For a stronger guarantee the blur would have to be baked into images on the server, a larger change that also removes the look-around.

4. Limits and what it does not do

- Dealer-chosen only: nothing is detected or blurred automatically (an automatic text blur is a possible later step).
- Areas can't be changed after the listing is live. An admin fix would need a backend change or moving the listing back to pending.
- Anyone can still look at the same street on Google Maps itself; this only controls what Plotraa shows.
- Moving the pin resets the areas (the view changes). The dealer sees the new view and marks them again.
- Frontend tests: npm test (6 projection tests). Backend: 124/124 pass (2 new tests for validation and the panorama match).

5. Deploy (order matters)

- Merge the street-view PRs first (backend #62, frontend #44). Then merge these; GitHub retargets them to main.
- Backend: `git pull && npm ci && npm run migrate && pm2 restart all --update-env`. **This one needs the migration.** No new env vars.
- Then redeploy the frontend (Vercel).

6. Test on a real phone after deploy

- Dealer preview (awaiting approval): the new section shows the locked street view; add, move, resize, remove and save boxes. Reload: the saved boxes are still there.
- Approve the listing, open the live page: the marked areas are blurred, everything else is sharp. Look left/right, up/down and zoom: the blur follows the sign.
- Try to save a change on a live listing (or after it went live in another tab): it is refused with a clear message.
- Move the pin on a pending listing: the marked areas are cleared and the dealer can mark them again.
- Address with no street imagery: no editor section and no street view, page loads normally.